

2018). Concerningly, studies such as Datta et al. (2015) and Lambrecht and Tucker (2019) have further shown that, beyond human bias, algorithms deployed in hiring systems can also encode discriminatory behavior that disadvantages particular demographic groups. In recent years, the rise of LLMs with their increasing adoption in hiring systems has prompted researchers to empirically evaluate whether emerging AI-driven hiring systems perpetuate similar forms of algorithmic discrimination. For instance, Veldanda et al. (2023) replicated Bertrand and Mullainathan’s correspondence experiment to investigate AI-driven hiring bias across gender, race, maternity status, pregnancy status, and political affiliation. Additionally, An et al. (2024) explored whether LLMs exhibit race- and gender-based discrimination through an experiment, in which LLMs write an email to a named job candidate about a hiring decision. Likewise, Nghiem et al. (2024) analyzed the extent to which LLMs exhibit bias toward applicants based on their first names when making employment recommendations.

At the same time, a broader interdisciplinary literature documents additional risks associated with the use of LLMs in organizational decision making, including hallucinations, systematic errors, and the reinforcement of social disparities (e.g., Magesh et al. 2024, Huang et al. 2025). These concerns have motivated extensive research on the ethical, legal, and governance implications of algorithmic hiring systems (e.g., Raghavan et al. 2020, Li et al. 2021, Hunkenschroer and Kriebitz 2023). For instance, Raghavan et al. (2020) study how firms that develop algorithmic pre-employment assessments build, validate, and manage bias in practice, highlighting gaps between technical capabilities, organizational incentives, and regulatory oversight.

Despite this growing attention to algorithmic fairness and governance, existing work largely evaluates hiring algorithms in isolation and focuses on disparities across protected attributes. Unlike prior work that centers on protected characteristics, the self-preference bias we study arises from interactions among algorithmic components within the hiring pipeline itself. By uncovering this interaction-induced bias, our study highlights an emergent risk in algorithmic hiring systems that is not directly addressed by existing fairness audits or regulatory frameworks.

More broadly, our findings inform ongoing debates on AI governance and responsible operations by demonstrating how seemingly neutral design choices, such as deploying the same or similar models for content generation and evaluation, can systematically distort how scarce evaluation capacity is allocated across candidates. This perspective underscores the need for regulatory and organizational approaches that account not only for input data and protected attributes, but also for the interaction structure of AI-driven operational processes.

2.4. LLMs as Judges in Decision Making

The concept of LLM-as-a-Judge, first introduced by Zheng et al. (2023), refers to the use of LLMs as automated evaluators that assess responses and assign scores, and has gained traction in AI

research as an efficient method for evaluating model performance without human intervention. However, emerging evidence suggests that LLM-based evaluation frameworks may introduce self-preference bias, where models disproportionately favor their own generated responses over those produced by humans or alternative models. For example, Zheng et al. (2023) examine the potential of this paradigm but also identified its limitations—several biases, including self-enhancement bias (self-preference bias), arise when LLMs are used to judge responses, such as those they generate themselves. While the results show that strong models like GPT-4 achieve over 80% agreement with human preferences on multi-turn conversations, they caution that LLM judges may still introduce systematic distortions, especially in high-stakes evaluation settings. Expanding on this line of inquiry, Xu et al. (2024) formally define self-preference bias and empirically revealed that it is widespread across popular LLMs and multiple tasks (e.g., translation, mathematical reasoning) through extensive experiments. They further demonstrate that the bias is amplified in self-refinement pipelines, which enhance fluency and coherence but reinforce the model’s preference for its own outputs. Building on these findings, Panickssery et al. (2024) investigate the underlying mechanisms of LLM self-preferencing and find that a model’s ability to recognize its own outputs, i.e., its self-recognition capability, contributed significantly to this bias. Specifically, LLMs such as GPT-4 and LLaMA 2 exhibit non-trivial self-recognition capabilities, and there is a strong positive correlation between a model’s self-recognition capability and the magnitude of its self-preference bias.

While prior work has established the existence of self-preference bias in LLM benchmarks, its broader consequences in business operations, particularly in high-stakes scenarios like algorithmic hiring, remain underexplored. This study builds on existing research by empirically measuring self-preference bias in algorithmic hiring processes. By extending the discussion into real-world decision-making contexts, this study provides critical insights for AI governance, ethical AI deployment, and the design of unbiased algorithmic hiring frameworks.

3. Definition and Measurement of AI Self-Preference Bias

In this section, we formally define AI self-preference bias and present empirical strategies to quantify the extent of this bias in the context of algorithmic hiring.

3.1. Definition of AI Self-Preference Bias

Building on recent work (Panickssery et al. 2024, Wataoka et al. 2024), we conceptualize AI self-preference bias as the tendency of an LLM to favor content it generated itself over content from other sources. This bias can manifest in two distinct forms:

1. **LLM-vs-Human Self-Preference:** The tendency of an LLM to prefer its own generated content over human-written content.

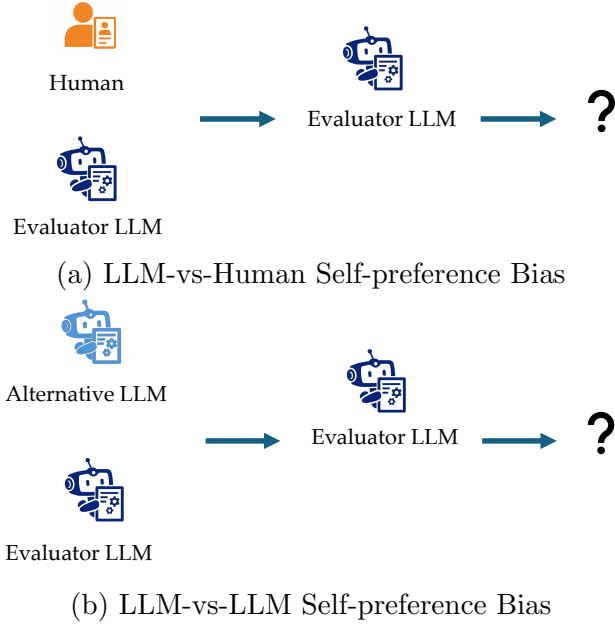


Figure 1 Illustration of the context of AI self-preference bias. In both cases, the evaluator LLM makes a choice between paired resumes, allowing us to test whether it systematically favors its own outputs.

2. LLM-vs-LLM Self-Preference: The tendency of an LLM to prefer its own output over content generated by an alternative LLM.

To empirically evaluate self-preference bias, we test whether a given LLM, acting as the evaluator (the evaluator LLM), makes unbiased selections when presented with carefully matched *pairs* of resumes (Figure 1). Each pair represents the *same underlying candidate*, with identical qualifications, experience, and background information, differing only in how that information is expressed. In the LLM-vs-human case (Figure 1a), the evaluator LLM compares a human-written resume to a counterfactual version it generated itself, with both describing the same candidate profile. In the LLM-vs-LLM case (Figure 1b), the evaluator LLM compares its own generated version to one produced by an alternative LLM. Under this design, the evaluator LLM’s task reduces to choosing between two alternative representations of the same candidate.

3.2. Quantifying AI Self-Preference Bias

When deployed in algorithmic hiring contexts, an LLM performing such pairwise comparisons effectively functions as a binary classifier. The measurement of bias in such classifiers has been extensively studied in the algorithmic fairness literature (e.g., Calders et al. 2009, Hardt et al. 2016). Building on this framework, we analyze both forms of self-preference through the lens of two foundational fairness criteria: *statistical parity* (Calders et al. 2009) and *equal opportunity* (Hardt et al. 2016), which capture distinct notions of fairness in algorithmic decision-making.